

NutrientNexus

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*A Digital Twin Autonomous System for Agricultural and Coastal Field
Monitoring and Decision Making*

Technical Insight

Problem

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- ① Crop growth
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Uncertainty.

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- Holistic, feedback- and context-driven, adaptive crop management system.
- Environment impact baked into the system.

Technical: unique selling point

Integrated Digital Twin System linking cause and effect across agriculture and water environments.

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Continuous feedback loop

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- Crop and soil data
- Crop growth history
- Environment contamination thresholds
- Profit margins?

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Weak link between farmers $\longleftrightarrow$ environment protection.

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Bridge gap between agriculture and marine protection.

Technical: SDG impact

Precision in fertiliser and irrigation application leads to

Metric	SDG indicator	Reasoning
Higher crop yield	2.3.1: Volume of production per labour unit	Decreased soil erosion & better water supply
More sustainable soil use	2.4.1: Proportion of sustainable agricultural area	Decreased soil erosion
Lower water eutrophication	14.1.1: Coastal eutrophication	Runoff-aware fertiliser application

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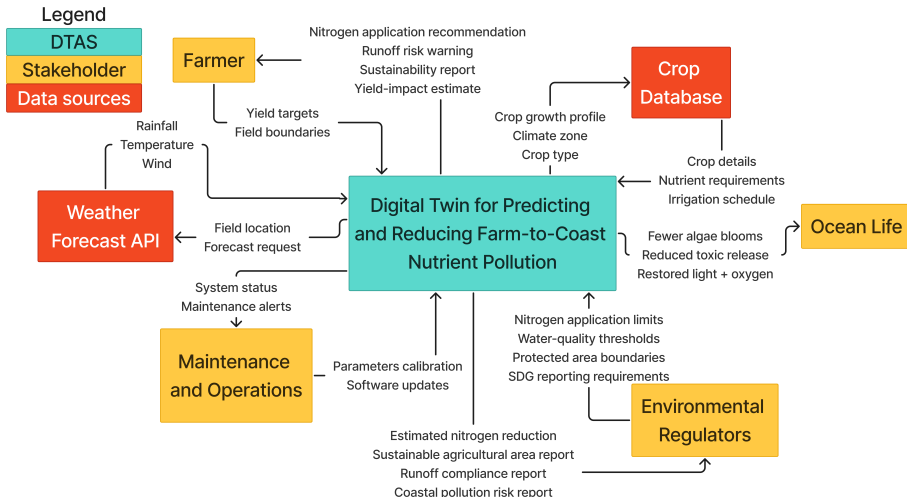
Who benefits?

Technical: SDG impact

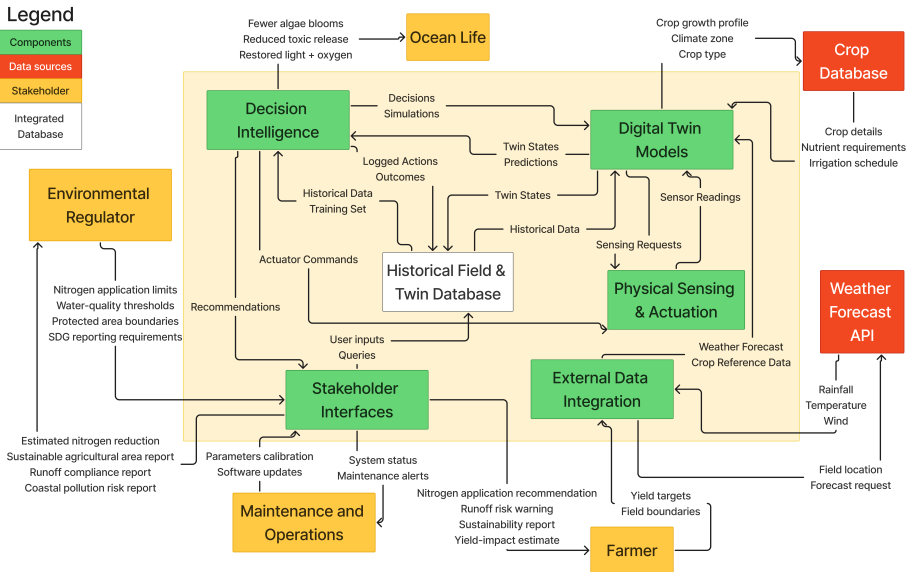
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Metric	Beneficiary	Reasoning
Higher crop yield	Customers & farmers	Higher supply → more affordable food; higher profit
Sustainable soil use	Natural habitats	Existing land reuse → less new land developed
Lower water eutrophication	Water ecosystems	Fewer algae blooms → more biodiversity & oxygen in water

Technical: level 0 context diagram



Technical: level 1 context diagram



Technical: concept map

"What challenges within either, or both SDGs 2 and 14, can be addressed by autonomous systems, and what role can digital twins play to meet these challenges?"

Wageningen University Smart Droplets Case Study

Texas A&M AgriLife Digital Twin Research

Cyber Agricultural/Coastal Systems

Environment contamin'n thresholds
History of crop growth
Per-crop nutrient/water needs
(Soil) water cycle model
(Soil) nutrient availability model
Water nutrients

Digital Twins

Agri/coastal UGV Swarms (Physical Twin)

Sensors

Precision robotic fert./sprinkles
Soil nutrients
Soil moisture
Thermal monitors
Plant transpiration

equipped with
guides
Plant identification camera

Cost/benefit agriculture-environment decision making

Precise fertiliser treatment

Precise irrigation strategies

Decreased erosion

Crop yield

Proportion of Sustainable Agricultural Area (Indicator 2.4.1) (Ref)

direct indicator of
Production Volume Per Labour Unit (Indicator 2.3.1) (Ref)

Agricultural nutrient runoff

Algae blooms

Less light/oxygen in water

Less plant growth

Ocean life

Forests and natural habitats

Consumers

Farmers

Long-term healthcare costs

Harm to humans

Longer land usage

Proportion of Sustainable Agricultural Area (Indicator 2.4.1)

Coastal Eutrophication Index (Indicator 14.1.1a)

Overuse of irrigation and fertilisers

Reduced habitat loss

Reduced spending for developing new land

Higher yield and profit

Tax income

Lower food prices

Government

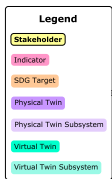
Double Smallholder Production (Target 2.3)

Production Volume Per Labour Unit (Indicator 2.3.1)

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Proof of Concept

Simplified system where one TurtleBot scans crops and generates and (potentially) applies fertiliser and irrigation recommendations.

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How exactly are we going to use digitally twinned TurtleBot to demonstrate the key aspects of our technical solution?

Proof of concept on the TurtleBot, concretely.

Physical entity: TurtleBot

- Navigates mapped crop zones.
- Provides simulated soil moisture, nutrient, and crop-growth data.
- Sends robot status: location, battery, and fertiliser stock.
- Simulates fertiliser application by rotating in place.

Digital twin entity

- Synchronises zone states after scans.
- Integrates weather, crop needs, and discharge thresholds.
- Computes recommendations *balancing* yield and runoff risk.
- Represents one robot now, but is designed for fleet extension.

PoC: DTAS characteristics

DTAS characteristic	How the PoC demonstrates it	Why it matters
Bidirectional communication	TurtleBot sends scan/status/action data; the digital twin returns recommendations and mission decisions.	Shows a closed feedback loop rather than passive monitoring.
State synchronisation	Zone readings, robot location, battery, fertiliser stock, scan status, and application events are mirrored digitally.	Proves the digital twin reflects the physical entity state.
Environment interaction	The robot moves through mapped crop zones and simulates fertiliser application by rotating in place.	Shows that the DTAS can connect recommendations to observable physical action.
Context-aware decision-making	The recommendation changes according to weather, crop needs, and runoff thresholds.	Shows that decisions depend on environmental context, not fixed rules only.

In scope	Out of scope
One TurtleBot as representative from future fleet	Full multi-robot coordination
Mapped crop-zone navigation	Real outdoor farming terrain
Stubbed moisture, nutrient, and crop growth readings	Real agricultural sensors
Rotation in place to simulate fertiliser application	Real fertiliser dosing, spreading, or irrigation
Simple context-based fertiliser & irrigation recommendations	Advanced agronomic/coastal ecosystem models
Simple stakeholder dashboard/report	Production-ready farmer or regulator platform

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The PoC proves the closed loop on one robot before scaling to a fleet.

PoC: functional requirements

ID	Requirement	Part	Why
FR1	Load mapped crop zones and show robot position	Digital/interface	Defines scan and action locations.
FR2	TurtleBot autonomously visits crop zones	Physical	Shows autonomous field coverage.
FR3	Return simulated moisture, nutrient, and growth data	Physical/sim	Replaces unavailable real sensors.
FR4	Synchronise zone and robot state with digital twin	Digital twin	Proves physical–digital alignment.

PoC: functional requirements (contd.)

ID	Requirement	Part	Why
FR5	Use weather, crop data, and runoff thresholds	Digital twin	Adds environmental context.
FR6	Compute fertiliser/irrigation recommendation	Decision module	Core agricultural decision-making.
FR7	Rotate in place to simulate fertiliser application	Physical	Shows recommendation becoming action.

PoC: non-functional requirements

ID	Requirement	Details	Reason
NFR1	Explainable advice	Show key decision/input factors	Builds farmer trust.
NFR2	Robust obstacle handling	Handle blocked-off paths	Robustness under different conditions.
NFR3	Modular software	Separate stubs, twin, decision, UI	Easy to replace with real sensors.
NFR4	Traceable logging	Store scans, recommendations, actions	Enables report and debugging.
NFR5	Safe actuation	Movement & actuation only when safe	Avoids robot damage & incorrect fertiliser/irrigation application.

Re-quire-ment	Demonstration	Expected result	Stakeholder relevance
FR1, FR2, NFR5	Show field map visit crop zones	Robot visits zones without collision	Shows autonomous crop-zone coverage.
FR3, FR4	Display stubbed readings and live DT updates	Zone states update after each scan	Shows physical-digital synchronisation.
FR5, FR6, NFR1	Change weather/threshold input	Irrigation/fertiliser advice changes	Shows context-aware decision making.
FR7	Robot performs fertiliser application	Robot rotates in place	Shows that a recommendation can lead to physical action.
FR5, FR6, NFR2	Generate sustainability report	Report includes commenced actions, any relevant amounts	Actual effect on eutrophication/soil sustainability.

Resource	Used for
TurtleBot	Navigation, telemetry, rotation action
ROS2 navigation/commands	Waypoints and in-place rotation
Gazebo/RViz	Simulation and visual debugging
Weather and crop data	Context for decision-making logic
Team members	Implementation of the PoC

Aspect	Viability	PoC consequence
Movement	Feasible in controlled lab/simulation.	Use mapped crop zones, not real terrain.
Fertiliser action	Real application is unavailable.	Use in-place rotation as visible proxy.
Scanning	Real sensors unavailable.	Use stubbed sensor values per zone.
Synchronisation	Feasible for one robot and few zones.	Log robot state and twin state updates.
Decision model	Full agronomy model too complex.	Use explainable rule-based thresholds.
Stakeholder output	Reports can be generated from logs.	Show farmer advice and runoff warning.

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PoC solution

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- Simulate fertiliser application by rotating the TurtleBot in place at the selected zone.

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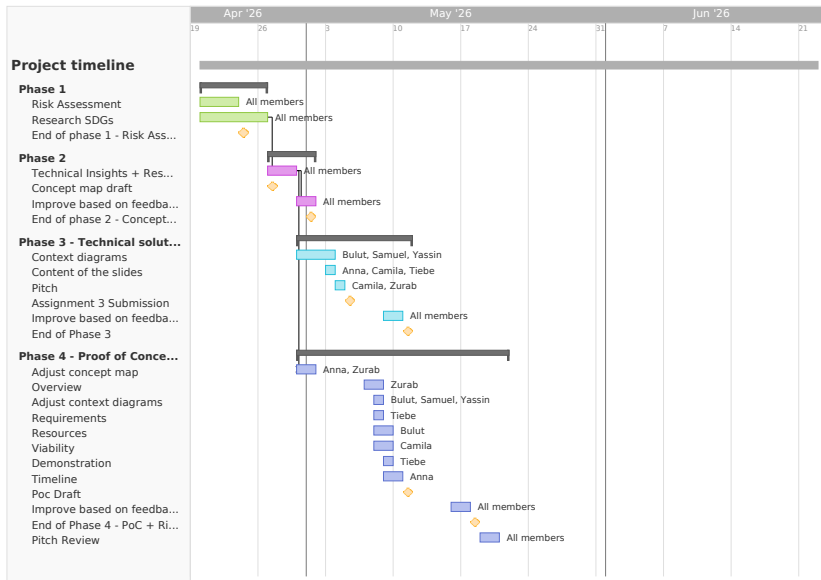
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Why this is still valid

The PoC tests the key aspect of technical solution — environment-based agricultural decision-making system.

scan → weigh in data → decide → act/report

PoC: timeline



PoC: timeline (contd.)

